

Retail Customer Retention Analytics – IKEA

Project Overview:

IKEA is a global leader in affordable, well-designed home furnishing products, operating through large-format stores, city stores, and a strong e-commerce presence. With increasing competition from online-first furniture brands, local retailers, and marketplaces, customer retention and repeat purchases have become critical for IKEA's long-term growth.

IKEA collects extensive data on customer purchases, loyalty memberships (IKEA Family), store visits, and online interactions. However, current reporting is largely descriptive and does not fully answer key business questions such as:

- Why are customers becoming inactive or churning?
- Which customers are loyal vs. at risk of churn?
- How effective are loyalty benefits, promotions, and omnichannel strategies?
- How do store formats and regions influence customer retention?

You are hired as a Power BI Analyst to design an IKEA Customer Retention Analytics Dashboard that integrates multiple datasets and delivers actionable, data-driven insights for business leaders.

Project Objective: Develop a robust, Interactive Customer Retention Analytics Dashboard in Power BI using the provided data, which will:

- Consolidate customer demographics, transaction history, store performance, and loyalty program usage.
- Enable dynamic segmentation of high-value, repeat, and churned customers.
- Offer actionable insights for improving retention, loyalty program effectiveness, and regional store strategies.

(If you're unfamiliar with concepts like churn rate, CLV, or other domain-specific terms, feel free to take the help of Google or AI tools like ChatGPT or Gemini to understand them better and apply them effectively in your project.)

Dataset Description: [Please Download dataset from this link \(all dataset\)](#)

1. **Customer_Demographics**: Contains demographic and membership details for each IKEA customer. Used to segment customers based on age, gender, region, income level, and membership tenure.
2. **Customer_Transactions**: Logs individual purchase transactions by customers across IKEA stores. Used for purchase behavior, frequency, and loyalty analysis.
3. **Store_Locations** : Contains metadata about IKEA store locations. Supports store-wise performance and retention analysis.
4. **Loyalty_Program** :Tracks customer participation in IKEA's loyalty program. Used for analyzing points behavior, tier effectiveness, and redemption trends.
5. **Churn_Labelled_Customers** : Provides a churn flag and behavioral insight for each customer based on the date of last purchase. Supports churn rate analysis and customer reactivation strategies.

Task to be performed:

(Please refrain from using AI to perform the tasks mentioned below, as it will only provide generic solutions.)

Task 1: Data Modeling and Cleaning (10 Marks)

Goal: Prepare the data for analysis

- Use Power Query Editor to load and transform the datasets.
 - Remove duplicate rows based on key columns (e.g., Customer_ID, Store_ID)
 - Format columns: Convert dates, ensure numeric types for Amount, Points, etc.
 - Handle missing or null values appropriately (e.g., filter or replace)
- Create calculated columns:
 - $\text{Membership_Duration} = \text{Today} - \text{Membership_Since}$
 - Add a Transaction_Year and Transaction_Month column from Transaction_Date
- Create a basic Data Model view

- One-to-Many: Customer_Demographics → Transactions, Loyalty_Program, Churn_Labelled_Customers
- Many-to-One: Transactions → Store_Locations

Task 2: Churn and Retention Metrics (10 Marks)

Goal: Identify churn trends across segments

- Create a Churn Rate card: $(\text{Churned Customers} / \text{Total Customers}) * 100$
- Visualize churn rate by:
 - Region
 - Income Group
 - Store Type (via store join)
 - Identify top 5 segments with highest churn %
- Create funnel chart: Total Customers → Repeat → Churned

Task 3: Repeat Purchase Analysis (10 Marks)

Goal: Understand customer loyalty through repeat behavior

- Create a measure:
 - Low-Tier Customers: (2-4 purchases)
 - Mid-Tier Customers: (5-10 purchases)
 - High-Tier Customers: (11+ purchases)
- Compare avg. purchase frequency by:
 - Region
 - Age Group
 - Loyalty Tier
- Identify product categories most frequently bought by loyal customers

Task 4: Promotion & Loyalty Impact (10 Marks)

Goal: Evaluate how promotions and loyalty tier affect retention

- Create Measure:
 - % of transactions where promotion was applied
 - Avg. purchase amount with vs without promotion
- Compare churn rate across loyalty tiers
- Show Points Earned vs Redeemed by Tier using clustered column chart
- Recommend how to improve redemption and retention

Task 5: Store Performance vs Retention (10 Marks)

Goal: Link store characteristics to retention outcomes

- Merge Store_ID from transactions with Store_Locations
- Visualize:
 - Avg. transaction amount by store type
 - Churn rate by store region
 - Correlation between store opening year and retention
- Suggest where to run store-specific campaigns

Task 6: Customer Value (CLV) Analysis (10 Marks)

Goal: Identify and prioritize high-value customers

Calculate $CLV = \text{Total Amount Spent} / \text{Membership Duration (in years)}$

- Segment customers into:
 - Low (Bottom 25%)
 - Medium (Mid 50%)
 - High (Top 25%)
- Visualize:
 - CLV vs Days Since Last Purchase
 - CLV by Loyalty Tier and Region

Task 7: Final Dashboard and Executive summary (20 Marks)

Goal: Communicate insights for business action

- Create a multi-page Power BI report:
 - Page 1: Overview KPIs (Churn, CLV, Repeat Rate)
 - Page 2: Loyalty & Promotion Impact
 - Page 3: Store and Region Insights
 - Page 4: Customer Segmentation (Churned, Repeat, High-Value)

Add slicers: Region, Income Group, Loyalty Tier, Store Type

- Create a card or textbox summarizing top 3 recommendations:
- What should IKEA do to retain more customers?
- Where should they focus next?

Task 8: Video explanation: Expressing the finding and actionable insights (20 Marks)

(The summary should be in your own words and must not be generated using AI. Please don't write a script and read it aloud. Also, screenshare and show the key findings. Marks will be deducted for failing to do so.)

Content: 3-5 minute video explaining:

- Introduction to the project.
- Problem statement.
- Key findings and insights.

Project Submission:

Note:

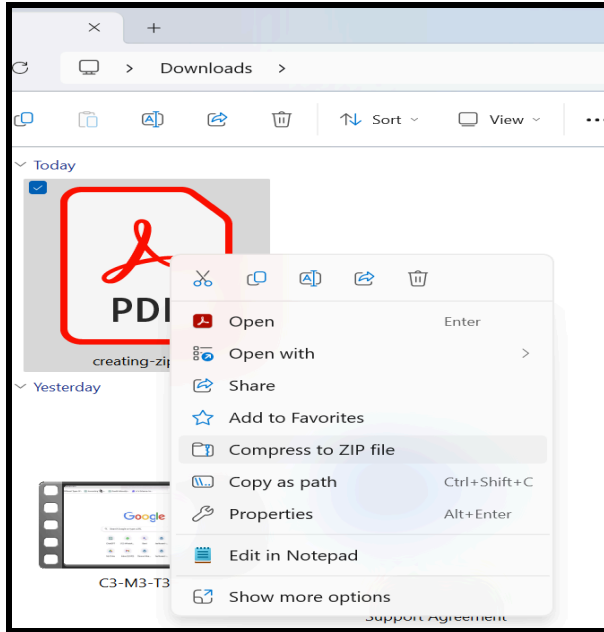
- The whole project should be submitted in a word file showing all tasks wherever with calculations, reporting features and dashboard with screenshots and the final submission should be in PDF file.
- Plagiarism will result in a penalty, including possible project disqualification.
- The project will be evaluated based on the quality of analysis and visualizations, depth of insights, feasibility of recommendations, clarity of explanations, and adherence to instructions and deliverables.

Submission Guidelines:

- Save the PDF in a folder and then convert it into a zipped (.zip) folder. **(Please note, the drivelink for the video created should also be added in the PDF itself.)**
- Upload the zipped folder on your respective dashboard.
- Failure to comply with submission guidelines will result in no grading/0 marks.

How to ZIP a PDF file:

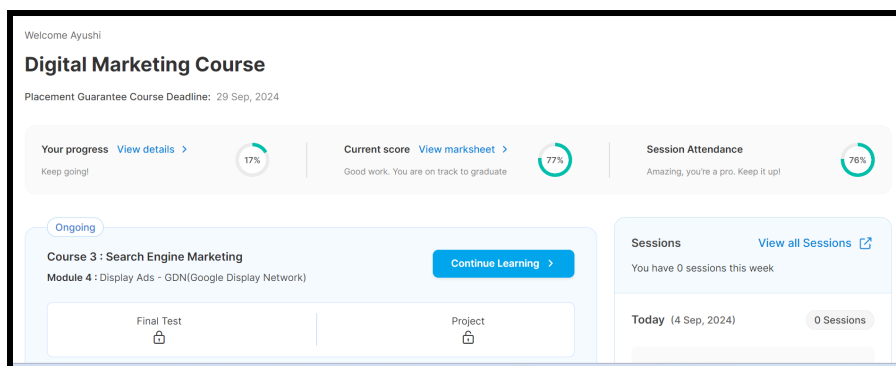
- Put all of the PDF you want to compress (or just one) into a new folder.
- Right click on that folder.



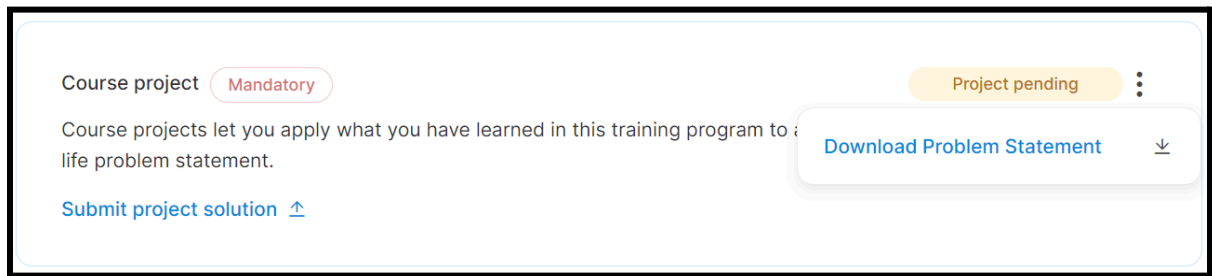
- Select the “Compress to ZIP file” option and then click “Compressed (Zipped) folder.”
- A new .ZIP file will be created that contains your document(s).

In order to submit the projects please follow the following steps:

1. Click on “Your progress [View details](#)” after logging into your dashboard.



- Next, click on the tab for the specific child course for which you want to download the problem statement. Then, scroll down to find the "**Course Project**" section.
- Now, click on the three dots on the right-hand side of the "Course Project" tab to select "**Download Problem Statement.**"



- Please follow the guidelines (screenshot is shared below) provided in the project to ensure correct submissions. Then, click on "**Upload Project Solution**" to submit your work.

